

Transformation Plan

AI Workforce Intelligence 2030

Phase 2 – Data Audit & Profiling

1. Objective

The purpose of this transformation plan is to identify any preprocessing or feature engineering activities that may be required before conducting Exploratory Data Analysis (EDA). Since the dataset has successfully passed the initial data quality assessment, only value-adding transformations are considered.

2. Mandatory Transformations

Based on data profiling and quality assessment, no mandatory transformations are required.

The dataset contains:

- No missing values
- No duplicate records
- Appropriate data types
- Consistent categorical values

Therefore, the dataset is considered analysis ready.

3. Optional Feature Engineering

The following transformations may be introduced during EDA if they improve business interpretation or support specific analytical objectives.

Proposed Transformation	Business Purpose
Experience Bands	Group employees into career stages (e.g., Entry, Mid-Level, Senior) for comparative analysis.
Salary Bands	Categorize salaries into Low, Medium, and High compensation groups to simplify workforce comparisons.
AI Risk Categories	Convert continuous AI Replacement Risk values into Low, Moderate, and High-risk segments for executive reporting.
Skills Standardization	Standardize or split skill values if multiple skills are stored in a single field.
Geographic Grouping	Group countries into regions if regional comparisons become necessary during analysis.

4. Transformation Strategy

Transformations will only be performed when they improve analytical clarity or help answer predefined business questions. No transformation will be introduced solely for technical purposes.

5. Conclusion

The dataset requires minimal preprocessing before analysis. Future transformations, if implemented, will focus on improving interpretability and supporting business storytelling rather than correcting data quality issues.